

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
4th Semester of B. Pharm (Old) Examination
University Theory Examination October/November 2016
PH210 Organic Chemistry-II

Date: 07/11/2016, Monday

Time: 01:30 p.m. to 04:30 p.m. Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1	Attempt any <u>TWO</u> of the following;	20
A	i. Explain Electrophilic Aromatic Substitution Reaction with mechanisms.	06
	ii. Explain criteria of aromaticity in details with huckel's rule.	04
B	What is green chemistry? Explain twelve principles of green chemistry.	10
C	i. Explain Friedal-craft acylation with reaction, mechanism and its limitation.	05
	ii. What are polynuclear aromatic compounds? Explain method of preparation and reaction of naphthalene	05
Q 2	Attempt any <u>TWO</u> of the following;	20
A	i. What is microwave? Differentiate Microwave Vs. Conventional Synthesis.	05
	ii. Write a short note on Nano chemistry and its application.	05
B	i. Write a short note on Neighbouring group effect with suitable example.	05
	ii. Explain Cycloaddition reaction with example.	05
C	i. Explain Sigmatropic reaction with example.	05
	ii. Write down preparations and reactions of Arenes.	05

SECTION- II

- 20 Q 3 Attempt any TWO of the following:
- 05 A i. Explain Hoffmann Rearrangement for Amide with suitable mechanism.
- 05 ii. Write down any three preparations and reactions of Amide.
- 05 B i. Explain in detail about preparation of Phenol.
- 05 ii. Write a note on Gabriel synthesis for primary amine.
- 05 C i. Explain in detail about preparation of Esters and Acyl chloride.
- 05 ii. Explain the Wolff Kishner Reduction with suitable Mechanism.
- 20 Q 4 Attempt any TWO of the following:
- 05 A i. Explain reaction and mechanism of Malonic ester synthesis.
- 05 ii. Explain Nucleophilic acyl substitution reaction.
- 05 B i. Explain the Wittig Reaction with suitable mechanism.
- 05 ii. Give any three preparations of Aldehyde and Ketone.
- 04 C i. Write a note on Kolbe synthesis with its mechanism.
- 06 ii. Give any three preparation and reaction of Carboxylic acid.